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Eduvos Assessment Coversheet

Academic Details

Campus: **Midrand**

Faculty: **Information Technology**

Module Code: **ITBDA4-14**

Lecturer's Name: **Ms Olorato Motitswane**

Assessment Details

Assessment Type: **Project**

Assignment/Brief Number: **2**

Due Date/ Submission Date: **13 November 2026**

Student Details (Individual, if applicable)

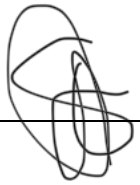
Student Full Name: **Gabriella Delgado**

Student Number: **EDUV4815041**

Academic Integrity

AI Disclosure Completed [X] **Yes** [ ] **No**

Declaration:
I declare that this assessment is my own original work except for source material explicitly acknowledged. I confirm that this work has not previously been submitted for assessments in any course. I acknowledge the institutional AI policy and confirm that AI tools have only been used as stipulated and documented in the AI Disclosure Appendix.

Student Signature 

Date **11 November 2026**

Lecturer Use Only

Lecturer Comments:

Marks Awarded (%):

Lecturer Signature:

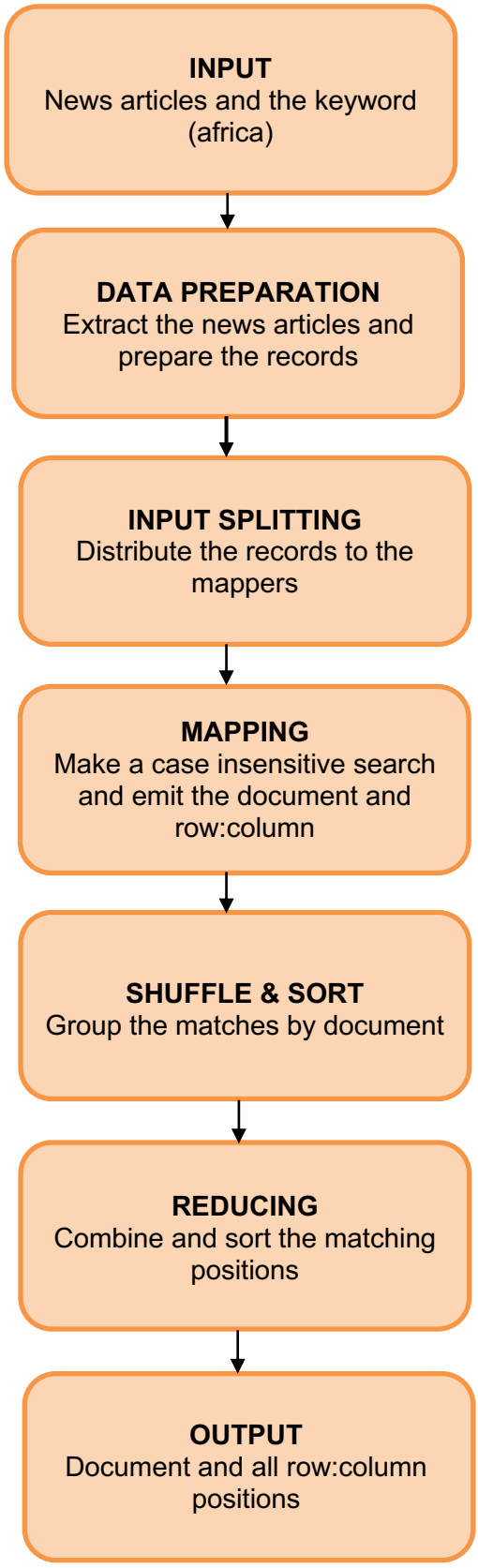
Date:

Assessment

QUESTION 1

QUESTION 1.1

Logical Layout Diagram of the MapReduce Framework with Keyword Search



Explanation of the above diagram:

The MapReduce framework will search all of the news articles from the news article corpus for a dynamically specified keyword such as “africa”. It will then identify and return the row and column positions in each document where the word is found (EMC Education Services, 2015). The process, which has been graphed above, is explained below:

1. **Input** – The news articles from the zip folder map_reduce_test_docs.zip are first extracted and used as the input dataset (EMC Education Services, 2015). The keyword is supplied dynamically when the program is executed.

2. **Data Preparation** – Each of the documents in the zip file which were extracted in the previous step are prepared into records which contain the filename, original row number and text in order to ensure that the keyword’s location can be easily and accurately found (EMC Education Services, 2015).

3. **Input Splitting** – Hadoop then divides the prepared dataset into what is called input splits and then distributes the records across multiple Mapper tasks in order for them to be processed parallel to one another (EMC Education Services, 2015).

4. **Mapping** – Each of the Mappers then performs a case insensitive search for the keyword (EMC Education Services, 2015). It identifies every time the word comes up and also its starting character column (EMC Education Services, 2015). For each match that is made, it produces a key value pair in the following format <document, row:column> (EMC Education Services, 2015).

5. **Shuffle & Sort** – Hadoop then automatically groups the key value pairs by document name in order to make sure that all the matching positions that belong to the same document are passed to the same Reducer (EMC Education Services, 2015).

6. **Reducing** – The reducer then collects and combines all the matching positions for each document and then sorts them numerically by row and column in order to produce one single output for each document (EMC Education Services, 2015).

7. **Output** – Lastly, the final results are then written to an output file which displays each document’s name and all of the row and column positions where the keyword occurs (EMC Education Services, 2015). Any document that doesn’t have a match is excluded.

QUESTION 1.2

The full code for Question 1.2 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 1.2. (File: MapReduce.R)

```
# Question 1.2 - MAPPER
map_keyword <- function(doc_name, row, text, keyword) {

  # Find all occurrences of the keyword, case insensitive
  columns <- gregexpr(tolower(keyword), tolower(text),
                      fixed = TRUE)[[1]]

  # Keep all valid positions
  columns <- columns[columns > 0]

  if (length(columns) == 0) return(list())

  # Match complete words only
  columns <- columns[sapply(columns, function(col) {
    before <- if (col == 1) "" else substr(text, col - 1, col - 1)
    after <- substr(text, col + nchar(keyword), col + nchar(keyword))

    !grepl("[[:alnum:]_]", before) &&
      !grepl("[[:alnum:]_]", after)
  })]

  # Produce key-value pairs
  lapply(columns, function(col) {
    list(key = doc_name, value = paste0(row, ":", col))
  })
}
```

12:33 prepare_articles(zip_file)

Console | Terminal | Background Jobs

R 4.5.2 · ~/Desktop/ITBDA_MapReduce/

```
> docs <- prepare_articles()
> map_keyword("doc1.txt", 1, docs[["doc1.txt"]][1], "africa")
[[1]]
[[1]]$key
[1] "doc1.txt"

[[1]]$value
[1] "1:64"
```

The above screenshot shows the Mapper function and how it finds the keyword and then produces a document - location key value pair for each time it is found.

QUESTION 1.3

The full code for Question 1.3 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 1.3. (File: MapReduce.R)

```
72
73  # QUESTION 1.3 - REDUCER
74  reduce_keyword <- function(key, values) {
75
76    rc <- do.call(rbind, strsplit(values, ":", fixed = TRUE))
77
78    values <- values[order(as.integer(rc[, 1]),
79                           as.integer(rc[, 2]))]
80
81    list(key = key, value = paste(values, collapse = ", "))
82  }
```

12:33 prepare_articles(zip_file)

Console | Terminal | Background Jobs

R 4.5.2 · ~/Desktop/ITBDA_MapReduce/

```
> reduce_keyword("doc2.txt", c("58:81", "1:57", "33:24"))
$key
[1] "doc2.txt"

$value
[1] "1:57, 33:24, 58:81"
```

The above screenshot shows the Reducer function and how it is able to sort and combine the keyword locations for a document. The function correctly sorts the positions numerically by row and column.

QUESTION 1.4

The full code for Question 1.4 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 1.4. (File: MapReduce.R)

```r
# Question 1.4 - Hadoop MapReduce Configuration
run_hadoop <- function(keyword, output = "output_africa") {

  # Validate the dynamically supplied keyword
  keyword <- trimws(keyword)
  if (!nzchar(keyword)) stop("Enter a keyword")

  # Prepare input data
  prepare_articles()

  # Locate Hadoop Streaming
  hadoop_home <- Sys.getenv("HADOOP_HOME")

  jar <- list.files(
    file.path(hadoop_home, "share/hadoop/tools/lib"),
    pattern = "^hadoop-streaming-.*\\.jar$",
    full.names = TRUE
  )

  if (length(jar) == 0) {
    stop("Hadoop Streaming JAR not found")
  }

  if (dir.exists(output)) {
    stop("Output folder already exists")
  }

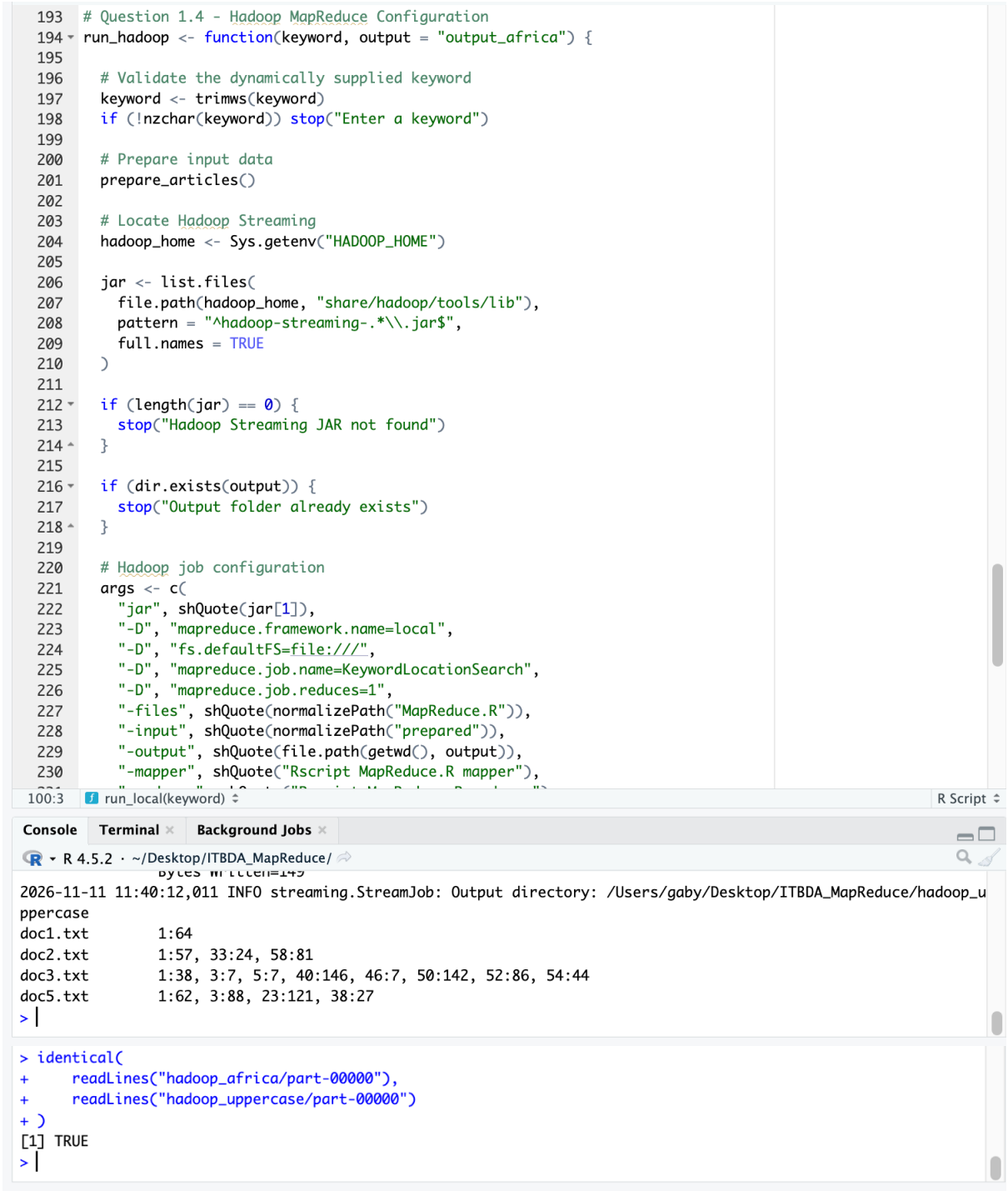
  # Hadoop job configuration
  args <- c(
    "jar", shQuote(jar[1]),
    "-D", "mapreduce.framework.name=local",
    "-D", "fs.defaultFS=file:///",
    "-D", "mapreduce.job.name=KeywordLocationSearch",
    "-D", "mapreduce.job.reduces=1",
    "-files", shQuote(normalizePath("MapReduce.R")),
    "-input", shQuote(normalizePath("prepared")),
    "-output", shQuote(file.path(getwd(), output)),
    "-mapper", shQuote("Rscript MapReduce.R mapper"),
```

12:33 prepare_articles(zip_file) R Script

Console Terminal Background Jobs

R 4.5.2 · ~/Desktop/ITBDA_MapReduce/

```
2026-11-11 11:36:16,616 INFO streaming.StreamJob: Output directory: /Users/gaby/Desktop/ITBDA_MapReduce/hadoop_a
frica
doc1.txt        1:64
doc2.txt        1:57, 33:24, 58:81
doc3.txt        1:38, 3:7, 5:7, 40:146, 46:7, 50:142, 52:86, 54:44
doc5.txt        1:62, 3:88, 23:121, 38:27
>
```



```
# Question 1.4 - Hadoop MapReduce Configuration
run_hadoop <- function(keyword, output = "output_africa") {

  # Validate the dynamically supplied keyword
  keyword <- trimws(keyword)
  if (!nzchar(keyword)) stop("Enter a keyword")

  # Prepare input data
  prepare_articles()

  # Locate Hadoop Streaming
  hadoop_home <- Sys.getenv("HADOOP_HOME")

  jar <- list.files(
    file.path(hadoop_home, "share/hadoop/tools/lib"),
    pattern = "^hadoop-streaming-.*\\.jar$",
    full.names = TRUE
  )

  if (length(jar) == 0) {
    stop("Hadoop Streaming JAR not found")
  }

  if (dir.exists(output)) {
    stop("Output folder already exists")
  }

  # Hadoop job configuration
  args <- c(
    "jar", shQuote(jar[1]),
    "-D", "mapreduce.framework.name=local",
    "-D", "fs.defaultFS=file:///",
    "-D", "mapreduce.job.name=KeywordLocationSearch",
    "-D", "mapreduce.job.reduces=1",
    "-files", shQuote(normalizePath("MapReduce.R")),
    "-input", shQuote(normalizePath("prepared")),
    "-output", shQuote(file.path(getwd(), output)),
    "-mapper", shQuote("Rscript MapReduce.R mapper"),
```

```
100:3  run_local(keyword)                                    R Script

Console  Terminal  Background Jobs

R 4.5.2 · ~/Desktop/ITBDA_MapReduce/

2026-11-11 11:40:12,011 INFO streaming.StreamJob: Output directory: /Users/gaby/Desktop/ITBDA_MapReduce/hadoop_u
ppercase
doc1.txt        1:64
doc2.txt        1:57, 33:24, 58:81
doc3.txt        1:38, 3:7, 5:7, 40:146, 46:7, 50:142, 52:86, 54:44
doc5.txt        1:62, 3:88, 23:121, 38:27
>
```

```
> identical(
+     readLines("hadoop_africa/part-00000"),
+     readLines("hadoop_uppercase/part-00000")
+ )
[1] TRUE
>
```

The above screenshot shows the implementation of the Hadoop MapReduce configuration. The first two screenshots show 1. The lower-case implementation with the keyword africa and 2. The upper-case implementation with the keyword AFRICA. The third screenshot shows that no matter the case of the keyword, the Hadoop configuration works and produces the same answer which confirms the configuration is case-insensitive. In the code, we have also included a local implementation of the MapReduce followed then by the Hadoop configuration. (File: MapReduce.R)

QUESTION 2

QUESTION 2.1

The full code for Question 2.1 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 2.1. (File: LDATopicModelling.R)

```r
# Question 2.1
#Data Preparation and Preprocessing

# Import the ZIP file
zip_file <- "/Users/gaby/Desktop/news.zip"

if (!file.exists(zip_file)) {
  stop("ZIP file not found")
}

# Extract the news articles
folder <- tempfile("news")
dir.create(folder)
unzip(zip_file, exdir = folder)

# Find all the news articles
files <- list.files(folder, pattern = "\\.txt$", recursive = TRUE, full.names = TRUE)

if (length(files) == 0) stop("No text files found")

# Import the articles into a data frame
news_df <- data.frame(
  file_name = basename(files),
  content = unname(vapply(files, function(file) {
    paste(readLines(file, warn = FALSE, encoding = "UTF-8"), collapse = " ")
  }, character(1))),
  stringsAsFactors = FALSE
)

# Preprocess and denoise the text
corpus <- VCorpus(VectorSource(news_df$content))

corpus <- tm_map(corpus, content_transformer(tolower))
corpus <- tm_map(corpus, removePunctuation)
corpus <- tm_map(corpus, removeNumbers)
corpus <- tm_map(corpus, removeWords, stopwords("en"))
corpus <- tm_map(corpus, stripWhitespace)

# Store the cleaned data in the data frame
news_df$content <- vapply(corpus, as.character, character(1))

# Display the data frame
str(news_df)
```

54:13 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.2 · ~/Desktop/ITBDA_MapReduce/

```
'data.frame':   10 obs. of  2 variables:
 $ file_name: chr  "news_1.txt" "news_10.txt" "news_2.txt" "news_3.txt" ...
 $ content  : chr  "eskom former official service provider appear court tuesday july eskom former employee servi
ce provider power u"| __truncated__ "wet cold weather expected continue western cape monday july cold wet weathe
r expected continue western cape wee"| __truncated__ "gauteng education commits implementing mayibuye primary sc
hool remedial actions tuesday july gauteng department"| __truncated__ "new labour minister charts way forward tu
esday july newly appointed employment labour minister nomakhosazana me"| __truncated__ ...
```

The above screenshot shows the articles being added to a data frame and then being preprocessed and the text denoised. The final and cleaned dataset is then displayed.

QUESTION 2.2

The full code for Question 2.2 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 2.2. (File: LDATopicModelling.R)

```r
# Question 2.2
# LDA Topic Modelling

# Create the Document Term Matrix
dtm <- DocumentTermMatrix(corpus, control = list(wordLengths = c(2, Inf)))

# Set the number of topics and top words
n_topics <- 5
k_words <- 10

# Create the LDA model
lda_model <- LDA(dtm, k = n_topics, control = list(seed = 1234))

# Display the top 10 words per topic
top_words <- terms(lda_model, k_words)
print(top_words)

# Obtain the topic word probability distributions
topic_probabilities <- posterior(lda_model)$terms

# Create a table of all the words and probabilities per topic
topic_words_df <- do.call(rbind, lapply(1:n_topics, function(i) {
  data.frame(
    Topic = paste("Topic", i),
    Word = top_words[, i],
    Probability = as.numeric(topic_probabilities[i, top_words[, i]])
  )
}))

rownames(topic_words_df) <- NULL

# Display the probability distribution of the top words
print(transform(topic_words_df, Probability = round(Probability, 4)))

```

85:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.2 · ~/Desktop/ITBDA_MapReduce/

```
> # Display the top 10 words per topic
> top_words <- terms(lda_model, k_words)
> print(top_words)
      Topic 1          Topic 2    Topic 3      Topic 4         Topic 5
 [1,] "invoices"       "eskom"    "police"     "said"          "said"
 [2,] "days"           "power"    "report"     "coalition"     "department"
 [3,] "public"         "court"    "sandf"      "government"    "cape"
 [4,] "said"           "service"  "school"     "public"        "unemployment"
 [5,] "departments"    "employee" "department" "governments"   "cold"
 [6,] "service"        "provider" "members"    "officials"     "meth"
 [7,] "paid"           "utility"  "said"       "role"          "priority"
 [8,] "gxoyiya"        "will"     "actions"    "policy"        "weather"
 [9,] "national"       "“eskom"   "four"       "psc"           "western"
[10,] "administration" "continue" "kidnappers" "relationships" "expected"
>
```

```r
# Question 2.2
# LDA Topic Modelling

# Create the Document Term Matrix
dtm <- DocumentTermMatrix(corpus, control = list(wordLengths = c(2, Inf)))

# Set the number of topics and top words
n_topics <- 5
k_words <- 10

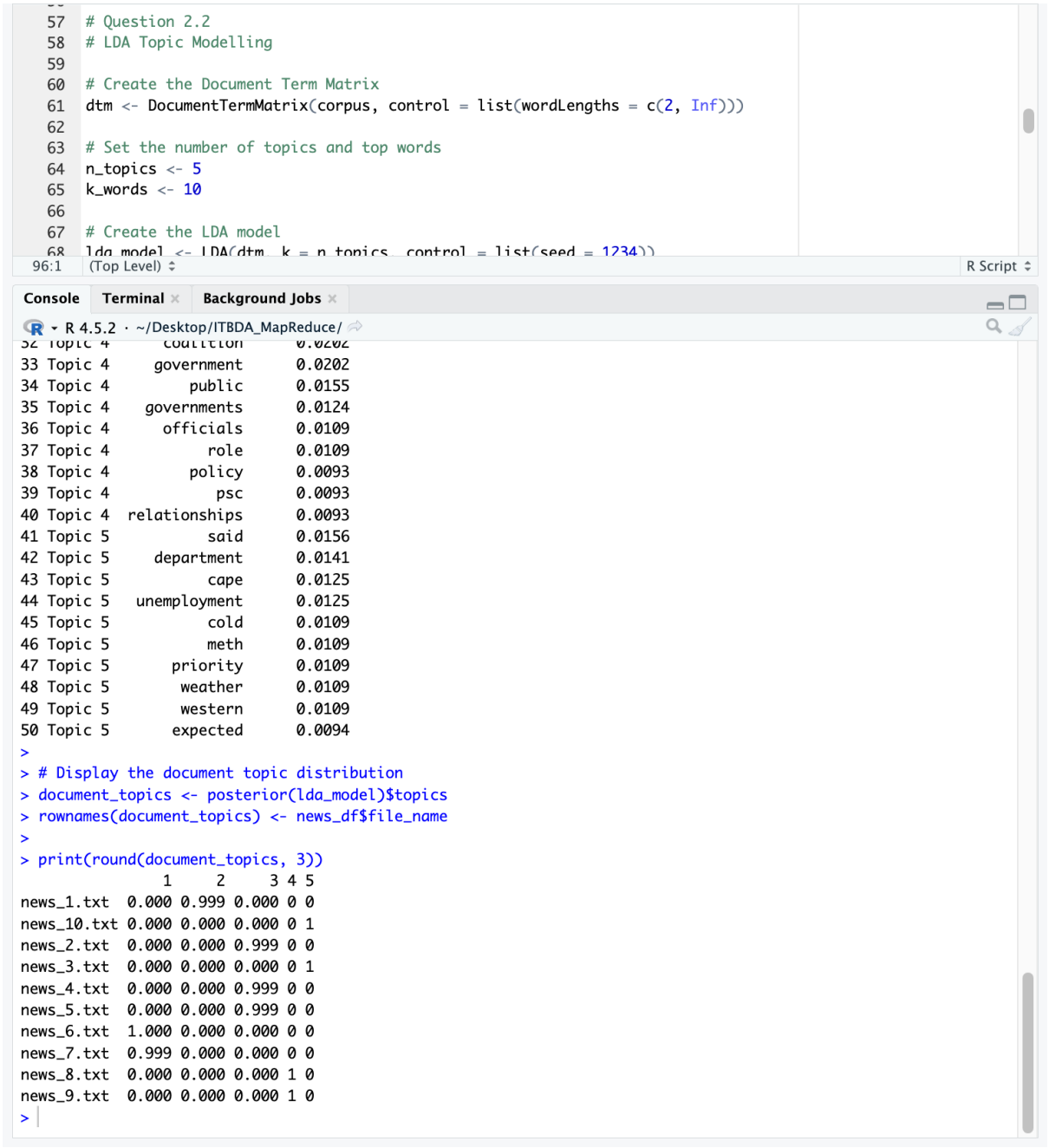
# Create the LDA model
lda_model <- LDA(dtm, k = n_topics, control = list(seed = 1234))
```

96:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.2 · ~/Desktop/ITBDA_MapReduce/

```
    Topic           Word Probability
1  Topic 1       invoices      0.0428
2  Topic 1           days      0.0277
3  Topic 1         public      0.0277
4  Topic 1           said      0.0277
5  Topic 1    departments      0.0252
6  Topic 1        service      0.0227
7  Topic 1           paid      0.0202
8  Topic 1        gxoyiya      0.0176
9  Topic 1       national      0.0176
10 Topic 1 administration      0.0151
11 Topic 2          eskom      0.0571
12 Topic 2          power      0.0357
13 Topic 2          court      0.0286
14 Topic 2        service      0.0286
15 Topic 2       employee      0.0214
16 Topic 2       provider      0.0214
17 Topic 2        utility      0.0214
18 Topic 2           will      0.0214
19 Topic 2         “eskom      0.0143
20 Topic 2       continue      0.0143
21 Topic 3         police      0.0229
22 Topic 3         report      0.0161
23 Topic 3          sandf      0.0138
24 Topic 3         school      0.0138
25 Topic 3     department      0.0115
26 Topic 3        members      0.0115
27 Topic 3           said      0.0115
28 Topic 3        actions      0.0092
29 Topic 3           four      0.0092
30 Topic 3     kidnappers      0.0092
31 Topic 4           said      0.0248
32 Topic 4      coalition      0.0202
33 Topic 4     government      0.0202
34 Topic 4         public      0.0155
35 Topic 4    governments      0.0124
36 Topic 4      officials      0.0109
```



```
57 # Question 2.2
58 # LDA Topic Modelling
59
60 # Create the Document Term Matrix
61 dtm <- DocumentTermMatrix(corpus, control = list(wordLengths = c(2, Inf)))
62
63 # Set the number of topics and top words
64 n_topics <- 5
65 k_words <- 10
66
67 # Create the LDA model
68 lda_model <- LDA(dtm, k = n_topics, control = list(seed = 1234))
```

96:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.2 · ~/Desktop/ITBDA_MapReduce/

```
32 Topic 4      coalition      0.0202
33 Topic 4     government      0.0202
34 Topic 4         public      0.0155
35 Topic 4    governments      0.0124
36 Topic 4      officials      0.0109
37 Topic 4           role      0.0109
38 Topic 4         policy      0.0093
39 Topic 4            psc      0.0093
40 Topic 4  relationships      0.0093
41 Topic 5           said      0.0156
42 Topic 5     department      0.0141
43 Topic 5           cape      0.0125
44 Topic 5   unemployment      0.0125
45 Topic 5           cold      0.0109
46 Topic 5           meth      0.0109
47 Topic 5       priority      0.0109
48 Topic 5        weather      0.0109
49 Topic 5        western      0.0109
50 Topic 5       expected      0.0094
>
> # Display the document topic distribution
> document_topics <- posterior(lda_model)$topics
> rownames(document_topics) <- news_df$file_name
>
> print(round(document_topics, 3))
               1     2     3 4 5
news_1.txt  0.000 0.999 0.000 0 0
news_10.txt 0.000 0.000 0.000 0 1
news_2.txt  0.000 0.000 0.999 0 0
news_3.txt  0.000 0.000 0.000 0 1
news_4.txt  0.000 0.000 0.999 0 0
news_5.txt  0.000 0.000 0.999 0 0
news_6.txt  1.000 0.000 0.000 0 0
news_7.txt  0.999 0.000 0.000 0 0
news_8.txt  0.000 0.000 0.000 1 0
news_9.txt  0.000 0.000 0.000 1 0
>
```

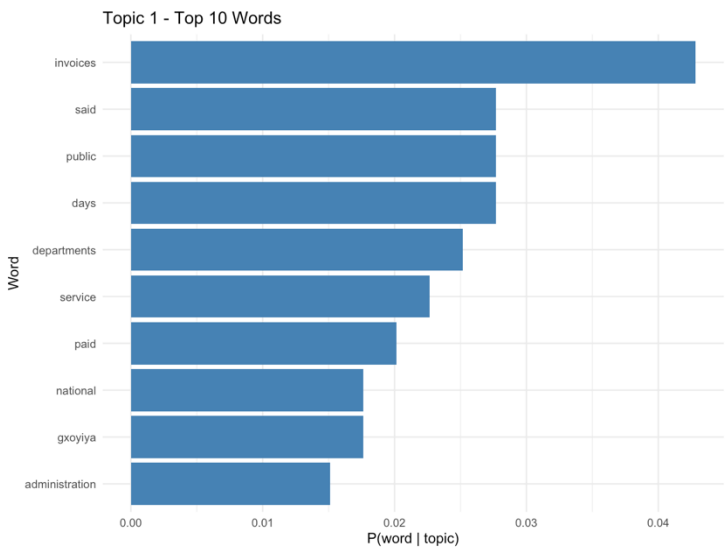
The above three screenshots show the output for the LDA topic modelling. The top 5 topics in the documents were determined and then the list of the top 10 words per topic were displayed (Screenshot 1). Screenshot 2 and 3 show all the words and their probabilities per topic as well as the last table showing the word distribution per topic. These probabilities represent the likelihood of that word occurring in the particular topic. The higher values mean that there is a stronger association between the word and the topic. The distributions help to see what the most representative words are which helps to interpret what the underlying themes of the news articles are.

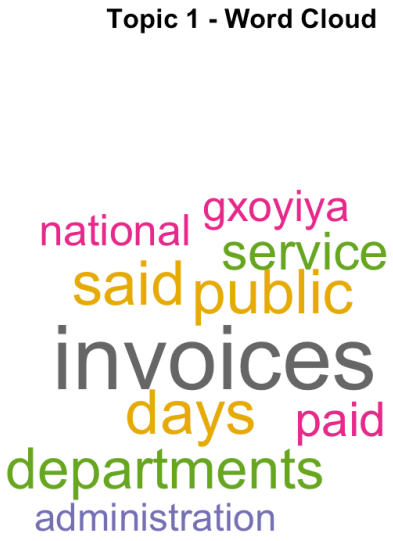
QUESTION 2.3

The full code for Question 2.3 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 2.3. (File: LDATopicModelling.R)

The following screenshots show two visualisations for each topic to show the top key words for each topic.

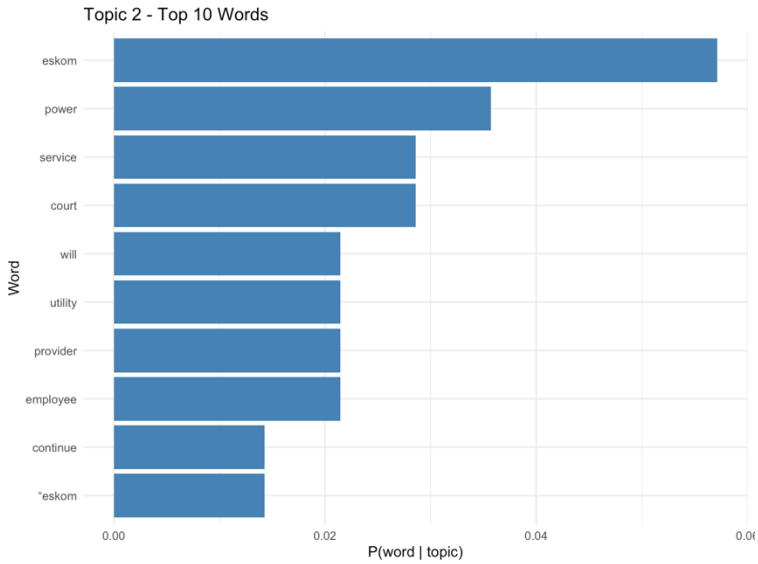
Topic 1





From the above we can see that some of topic 1's most dominant words are "invoices", "public" and "department" which could mean that this topic is largely about public admin and payments.

Topic 2

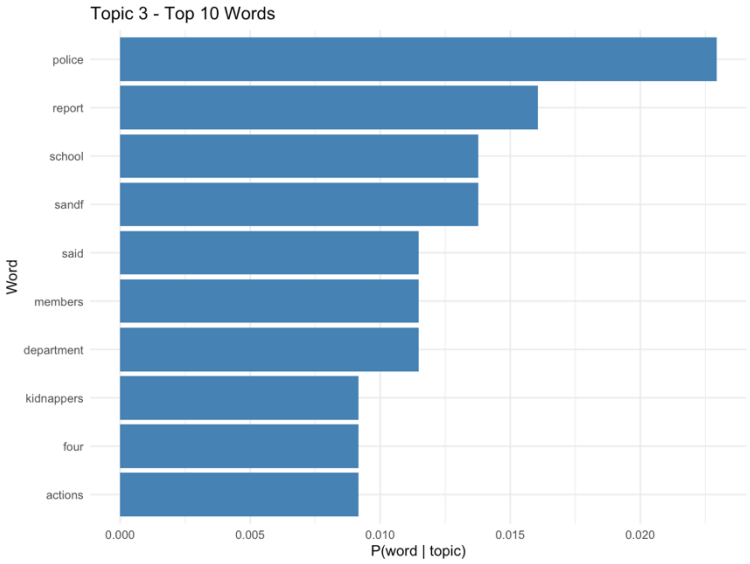




From the above we can see that some of topic 2's most dominant words are "eskom",

"power" and "service" which could mean that this topic is largely about electricity and any related services.

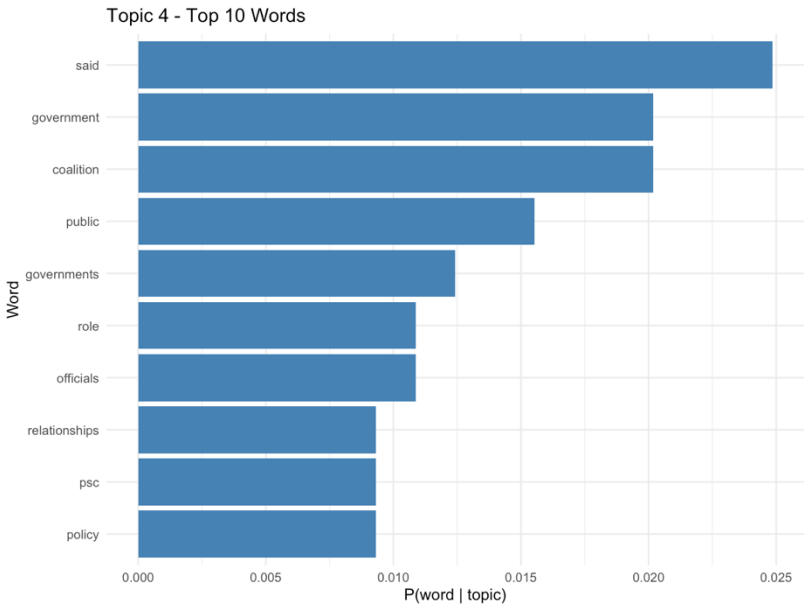
Topic 3





From the above we can see that some of topic 3's most dominant words are "police", "report", "school" and "kidnappers" which could mean that this topic is largely about crime near schools especially kidnappings.

Topic 4

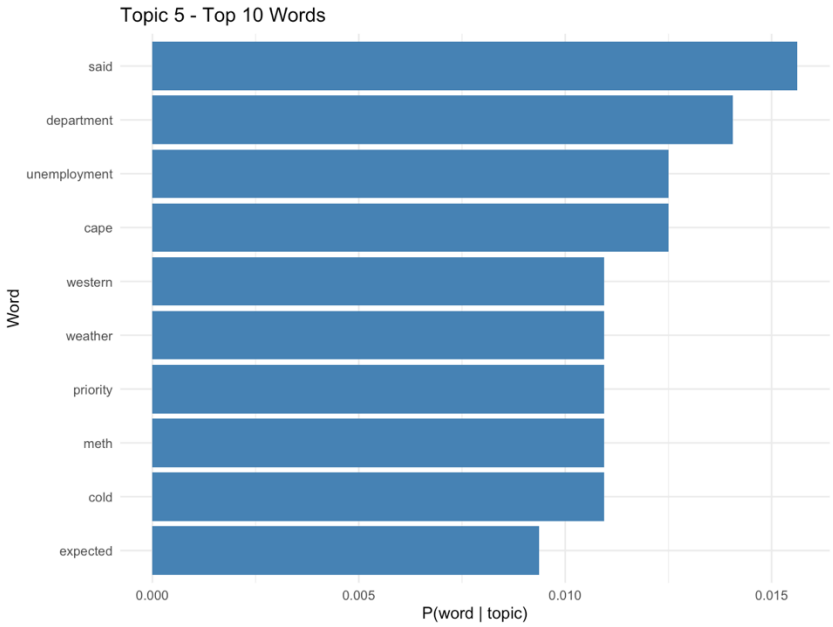


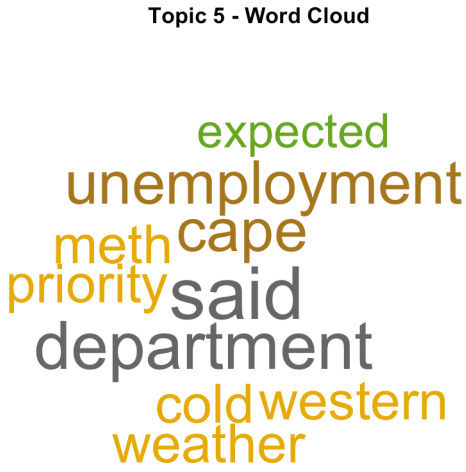


From the above we can see that some of topic 4's most dominant words are "government", "coalition", "officials" and "policy" which could mean that this topic is largely about political

affairs and government coalitions.

Topic 5





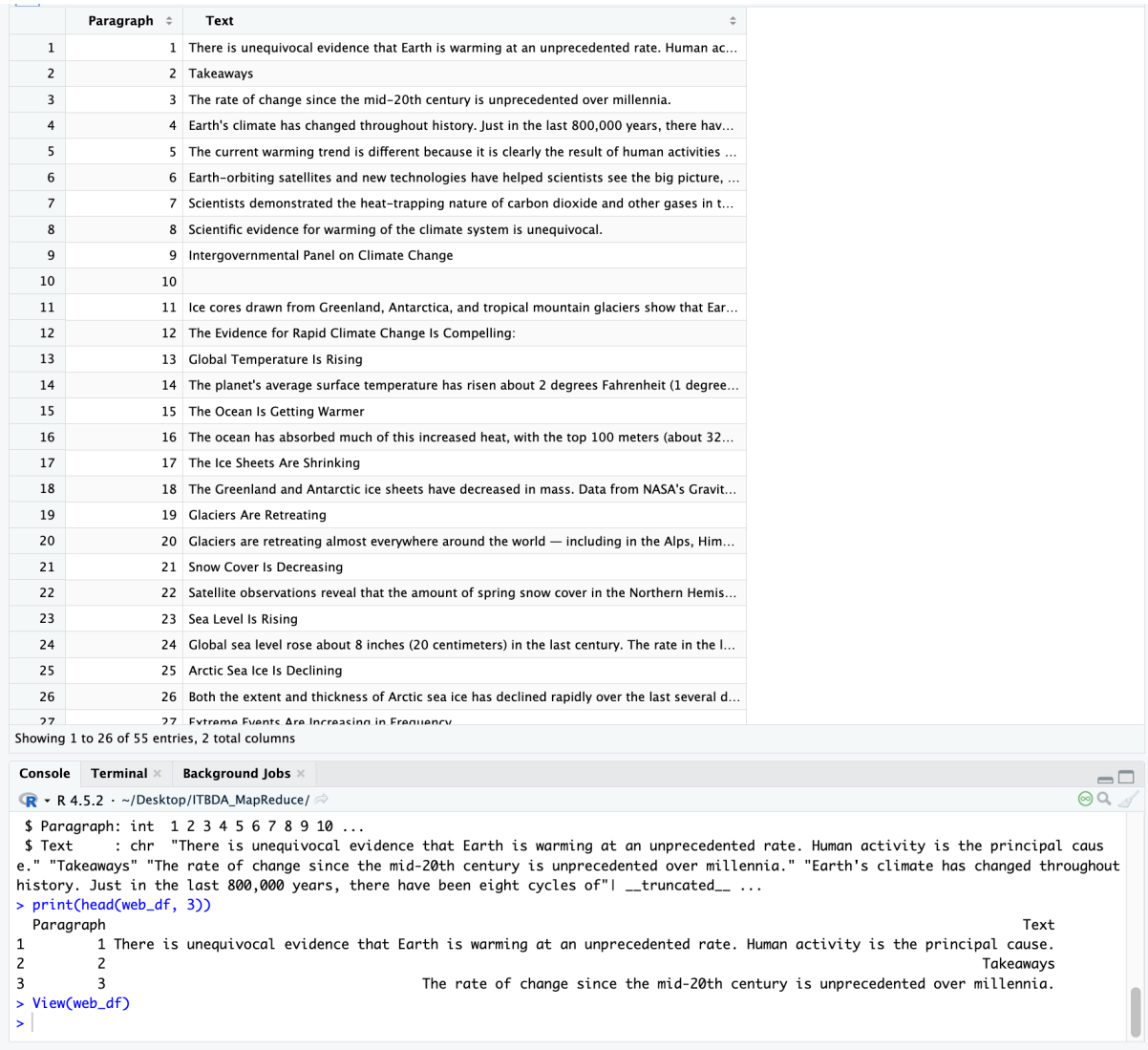
From the above we can see that some of topic 5's most dominant words are "department", "unemployment", "cape" and "weather" which could mean that this topic is largely about a mixture of employment issues and weather related issues in the Western Cape.

QUESTION 3

Website Used: https://science.nasa.gov/climate-change/evidence/

QUESTION 3.1

The full code for Question 3.1 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 3.1. (File: Q3WebScraping.R)



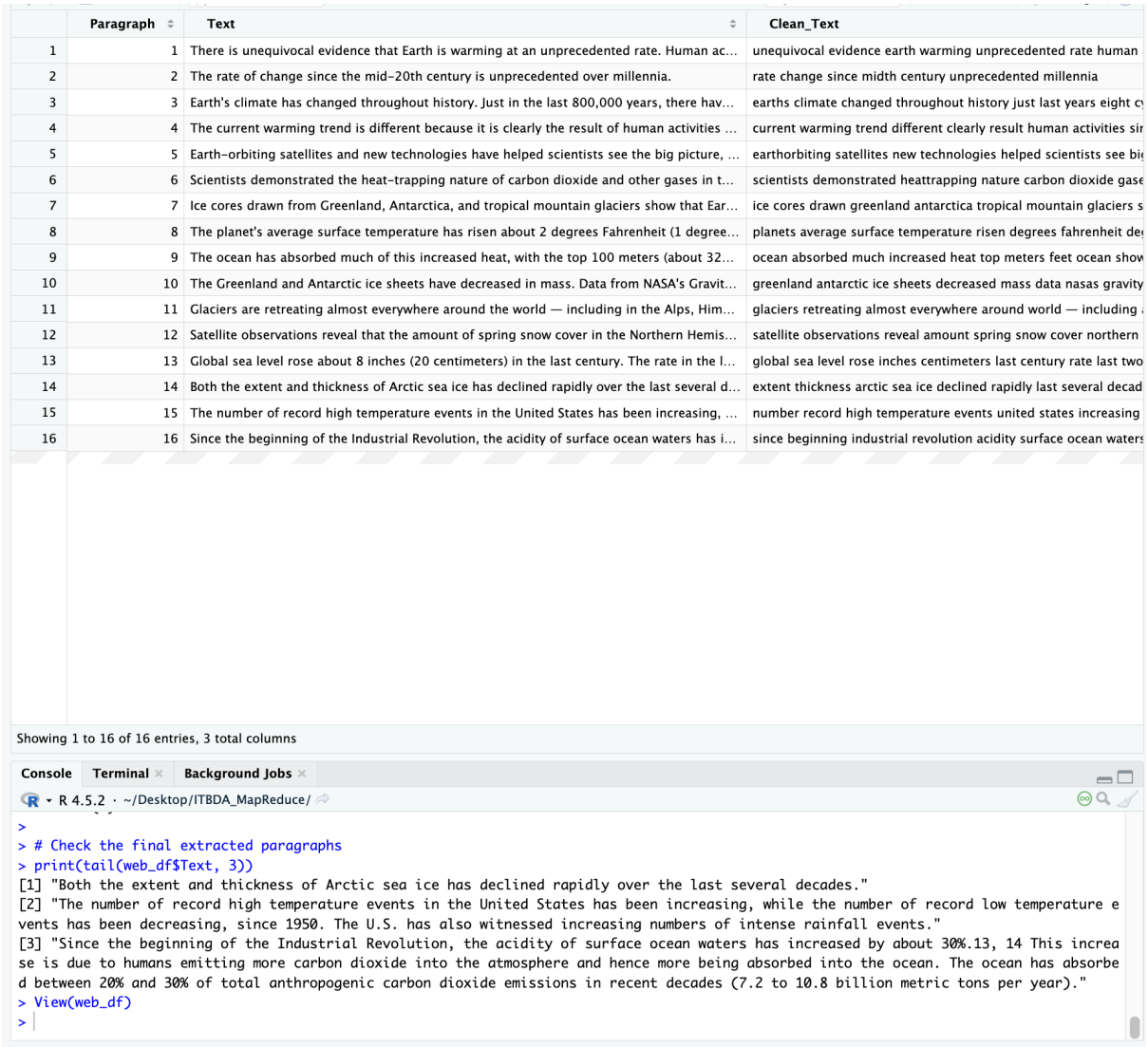
	Paragraph	Text
1	1	There is unequivocal evidence that Earth is warming at an unprecedented rate. Human ac…
2	2	Takeaways
3	3	The rate of change since the mid-20th century is unprecedented over millennia.
4	4	Earth's climate has changed throughout history. Just in the last 800,000 years, there hav…
5	5	The current warming trend is different because it is clearly the result of human activities …
6	6	Earth-orbiting satellites and new technologies have helped scientists see the big picture, …
7	7	Scientists demonstrated the heat-trapping nature of carbon dioxide and other gases in t…
8	8	Scientific evidence for warming of the climate system is unequivocal.
9	9	Intergovernmental Panel on Climate Change
10	10	
11	11	Ice cores drawn from Greenland, Antarctica, and tropical mountain glaciers show that Ear…
12	12	The Evidence for Rapid Climate Change Is Compelling:
13	13	Global Temperature Is Rising
14	14	The planet's average surface temperature has risen about 2 degrees Fahrenheit (1 degree…
15	15	The Ocean Is Getting Warmer
16	16	The ocean has absorbed much of this increased heat, with the top 100 meters (about 32…
17	17	The Ice Sheets Are Shrinking
18	18	The Greenland and Antarctic ice sheets have decreased in mass. Data from NASA's Gravit…
19	19	Glaciers Are Retreating
20	20	Glaciers are retreating almost everywhere around the world — including in the Alps, Him…
21	21	Snow Cover Is Decreasing
22	22	Satellite observations reveal that the amount of spring snow cover in the Northern Hemis…
23	23	Sea Level Is Rising
24	24	Global sea level rose about 8 inches (20 centimeters) in the last century. The rate in the l…
25	25	Arctic Sea Ice Is Declining
26	26	Both the extent and thickness of Arctic sea ice has declined rapidly over the last several d…
27	27	Extreme Events Are Increasing in Frequency

Showing 1 to 26 of 55 entries, 2 total columns

The above screenshot shows evidence that we extracted the text from the NASA webpage. The website that was used is: https://science.nasa.gov/climate-change/evidence/. The extracted text was then stored in a data frame called web_df, which is displayed above. The data frame has the paragraph number and its corresponding text.

QUESTION 3.2

The full code for Question 3.2 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 3.2. (File: Q3WebScraping.R)



	Paragraph	Text	Clean_Text
1	1	There is unequivocal evidence that Earth is warming at an unprecedented rate. Human ac...	unequivocal evidence earth warming unprecedented rate human
2	2	The rate of change since the mid-20th century is unprecedented over millennia.	rate change since midth century unprecedented millennia
3	3	Earth's climate has changed throughout history. Just in the last 800,000 years, there hav...	earths climate changed throughout history just last years eight cy
4	4	The current warming trend is different because it is clearly the result of human activities ...	current warming trend different clearly result human activities sir
5	5	Earth-orbiting satellites and new technologies have helped scientists see the big picture, ...	earthorbiting satellites new technologies helped scientists see bi
6	6	Scientists demonstrated the heat-trapping nature of carbon dioxide and other gases in t...	scientists demonstrated heattrapping nature carbon dioxide gase
7	7	Ice cores drawn from Greenland, Antarctica, and tropical mountain glaciers show that Ear...	ice cores drawn greenland antarctica tropical mountain glaciers s
8	8	The planet's average surface temperature has risen about 2 degrees Fahrenheit (1 degree...	planets average surface temperature risen degrees fahrenheit de
9	9	The ocean has absorbed much of this increased heat, with the top 100 meters (about 32...	ocean absorbed much increased heat top meters feet ocean show
10	10	The Greenland and Antarctic ice sheets have decreased in mass. Data from NASA's Gravit...	greenland antarctic ice sheets decreased mass data nasas gravity
11	11	Glaciers are retreating almost everywhere around the world — including in the Alps, Him...	glaciers retreating almost everywhere around world — including
12	12	Satellite observations reveal that the amount of spring snow cover in the Northern Hemis...	satellite observations reveal amount spring snow cover northern
13	13	Global sea level rose about 8 inches (20 centimeters) in the last century. The rate in the l...	global sea level rose inches centimeters last century rate last two
14	14	Both the extent and thickness of Arctic sea ice has declined rapidly over the last several d...	extent thickness arctic sea ice declined rapidly last several decad
15	15	The number of record high temperature events in the United States has been increasing, ...	number record high temperature events united states increasing
16	16	Since the beginning of the Industrial Revolution, the acidity of surface ocean waters has i...	since beginning industrial revolution acidity surface ocean waters

Showing 1 to 16 of 16 entries, 3 total columns

```
Console  Terminal  Background Jobs
R 4.5.2 · ~/Desktop/ITBDA_MapReduce/
>
> # Check the final extracted paragraphs
> print(tail(web_df$Text, 3))
[1] "Both the extent and thickness of Arctic sea ice has declined rapidly over the last several decades."
[2] "The number of record high temperature events in the United States has been increasing, while the number of record low temperature e
vents has been decreasing, since 1950. The U.S. has also witnessed increasing numbers of intense rainfall events."
[3] "Since the beginning of the Industrial Revolution, the acidity of surface ocean waters has increased by about 30%.13, 14 This increa
se is due to humans emitting more carbon dioxide into the atmosphere and hence more being absorbed into the ocean. The ocean has absorbe
d between 20% and 30% of total anthropogenic carbon dioxide emissions in recent decades (7.2 to 10.8 billion metric tons per year)."
> View(web_df)
>
```

The above screenshot shows evidence that the data frame with the web pages text has now been cleaned. The screenshot shows the original text and then the cleaned text which was preprocessed in order to remove any irrelevant information and also standardise the text. Any empty, short, duplicated and also credit related paragraphs as well as references were removed. The text normalisation included making the text lower case, removing punctuation, removing any numbers, removing English stop-words and also any excess whitespaces. The cleaned text was then stored separately from the original content. We then also created a Document Term Matrix in order to see the word frequencies which excluded any terms shorter than three characters which is needed for the TF-IDF and LDA analysis in the next question.

QUESTION 3.3

The full code for Question 3.3 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 3.3. (File: Q3WebScraping.R)

The two text summarisation techniques used in this question are:

1. TF-IDF – Is used to be able to identify the most informative terms (EMC Education Services, 2015). It is used to rank and then extract the important sentences which helps provide a good text summary (EMC Education Services, 2015).

2. LDA Topic Modelling – Is used to identify any underlying themes and then select the sentences that represent those themes (EMC Education Services, 2015). It is helpful for text summarisation as it offers a different approach to summarising the text from the above technique.

The two techniques are used together in order to provide a very well rounded text summary.

Technique 1 – TF IDF

Technique 1 which uses TF IDF identifies the following as a 5 sentence summary:

```
TF-IDF EXTRACT SUMMARY:
> cat(paste(sentences[tfidf_selected], collapse = " "), "\n")
Earth-orbiting satellites and new technologies have helped scientists see the big picture, collecting many different types of informatio
n about our planet and its climate all over the world. Ancient evidence can also be found in tree rings, ocean sediments, coral reefs, a
nd layers of sedimentary rocks. Glaciers are retreating almost everywhere around the world – including in the Alps, Himalayas, Andes, Ro
ckies, Alaska, and Africa.8

Image: Miles Glacier, Alaska Satellite observations reveal that the amount of spring snow cover in the Northern Hemisphere has decreased
over the past five decades and the snow is melting earlier. has also witnessed increasing numbers of intense rainfall events.
>
```

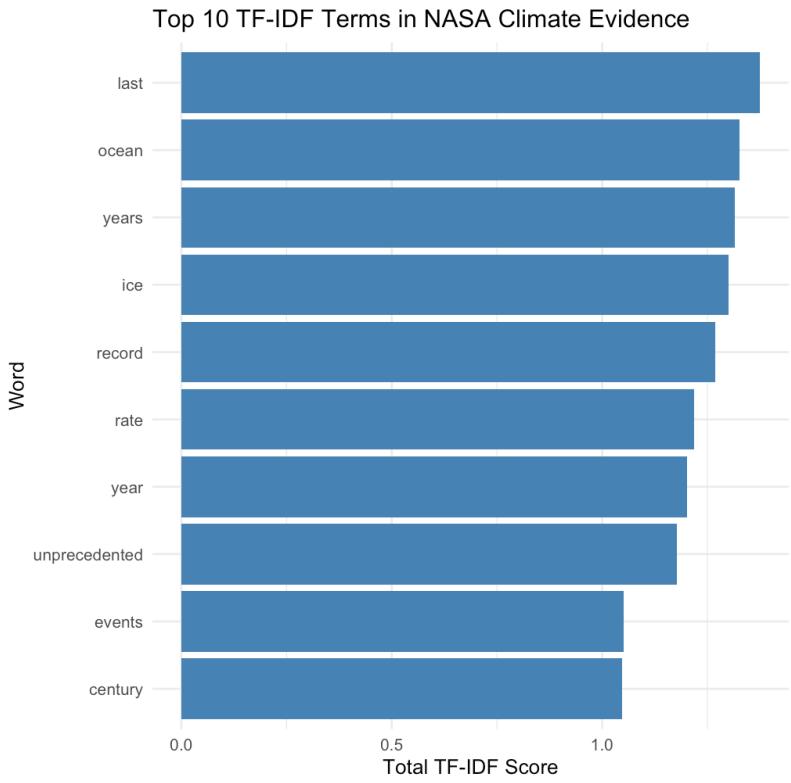
Technique 2 – LDA

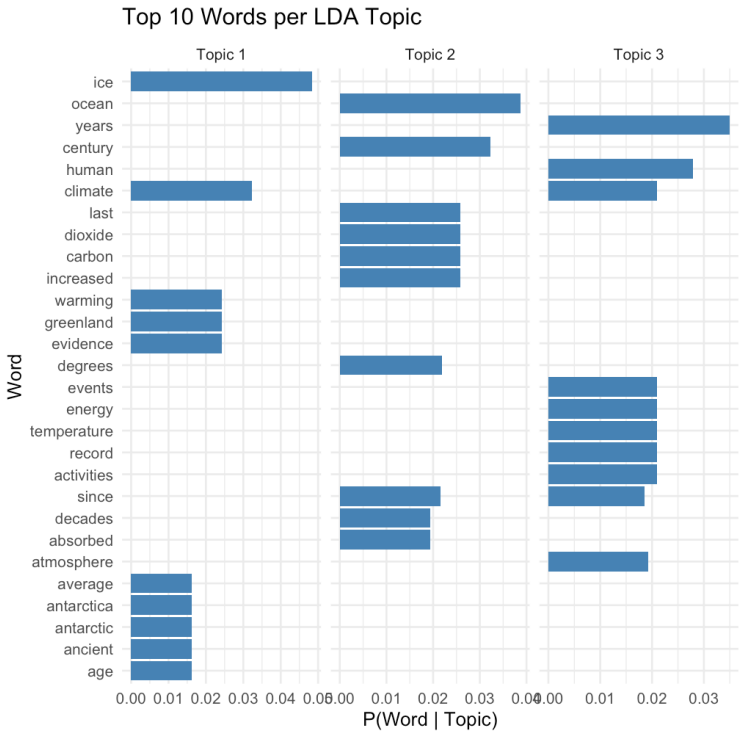
Technique 2 which uses LDA Topic Modelling identifies the following as a 5 sentence summary:

```
LDA TOPIC SENTENCE SUMMARY:
> cat(paste(sentences[chosen], collapse = " "), "\n")
This ancient, or paleoclimate, evidence reveals that current warming is occurring roughly 10 times faster than the average rate of warmi
ng after an ice age. The years 2016 and 2020 are tied for the warmest year on record. The ocean has absorbed much of this increased hea
t, with the top 100 meters (about 328 feet) of ocean showing warming of 0.67 degrees Fahrenheit (0.33 degrees Celsius) since 1969.6 Eart
h stores 90% of the extra energy in the ocean. The Greenland and Antarctic ice sheets have decreased in mass. Data from NASA's Gravity R
ecovery and Climate Experiment show Greenland lost an average of 279 billion tons of ice per year between 1993 and 2019, while Antarctic
a lost about 148 billion tons of ice per year.7

Image: The Antarctic Peninsula,
```

The following are the two visualisations used to represent the text summarisation based on the two techniques used above:





Overall Summary for Question 3.3

From the above, we can see that both the TF-IDF and LDA topic summaries successfully extracted the key information from NASA's climate change evidence webpage. These two techniques provided summaries that highlighted a main message which can be described by the following: accelerated global warming, record temperatures, ocean warming, glacier retreat, declining snow cover, ice sheet losses and even increasing extreme rainfall events. They also identified the importance of satellite observations as well as historical climate records in providing the evidence of climate change.

The two visualisations did support these findings with TF-IDF being able to highlight the informative terms such as “ocean”, “ice”, “record” and “rate” and LDA identifying the three related themes which are: climate warming and ice loss, ocean changes and carbon dioxide absorption and climate trends and human activity.

Both of the techniques did an extractive summary of the web page by being able to select five original sentences in order to preserve the source information. However, we can also see that the TF-IDF technique prioritised observational and historical evidence and LDA provided a more broader thematic coverage and more quantitative findings which included warming rates and ice sheet losses. From the above, one can get a good idea of a text summary of the web page.

QUESTION 4

Website used: https://www.scrapethissite.com/pages/simple/

QUESTION 4.1

The full code for Question 4.1 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 4.1. (File: Q4WebScraping.R)

	Country_Raw	Capital_Raw	Population_Raw	Area_Raw
1	Andorra	Andorra la Vella	84000	468.0
2	United Arab Emirates	Abu Dhabi	4975593	82880.0
3	Afghanistan	Kabul	29121286	647500.0
4	Antigua and Barbuda	St. John's	86754	443.0
5	Anguilla	The Valley	13254	102.0
6	Albania	Tirana	2986952	28748.0
7	Armenia	Yerevan	2968000	29800.0
8	Angola	Luanda	13068161	1246700.0
9	Antarctica	None	0	1.4E7
10	Argentina	Buenos Aires	41343201	2766890.0
11	American Samoa	Pago Pago	57881	199.0
12	Austria	Vienna	8205000	83858.0
13	Australia	Canberra	21515754	7686850.0
14	Aruba	Oranjestad	71566	193.0
15	Åland	Mariehamn	26711	1580.0
16	Azerbaijan	Baku	8303512	86600.0
17	Bosnia and Herzegovina	Sarajevo	4590000	51129.0
18	Barbados	Bridgetown	285653	431.0
19	Bangladesh	Dhaka	156118464	144000.0
20	Belgium	Brussels	10403000	30510.0
21	Burkina Faso	Ouagadougou	16241811	274200.0
22	Bulgaria	Sofia	7148785	110910.0

Showing 1 to 22 of 250 entries, 6 total columns

The above screenshot shows evidence of the country data from the web page: https://www.scrapethissite.com/pages/simple/. We constructed the data frame containing textual and numerical attributes. The extracted data initially contain spurious, inconsistent or incomplete information as requested. The next question will take care of data cleaning, preprocessing and denoising and will provide evidence of this.

QUESTION 4.2

The full code for Question 4.2 is in the R file which has been submitted as well. Please refer to the R file for the full answer to Question 4.2. (File: Q4WebScraping.R)

	country	capital	population	area_km2
1	Andorra	Andorra la Vella	84000	468.00
2	United Arab Emirates	Abu Dhabi	4975593	82880.00
3	Afghanistan	Kabul	29121286	647500.00
4	Antigua and Barbuda	St. John's	86754	443.00
5	Anguilla	The Valley	13254	102.00
6	Albania	Tirana	2986952	28748.00
7	Armenia	Yerevan	2968000	29800.00
8	Angola	Luanda	13068161	1246700.00
9	Antarctica	Not provided	0	14000000.00
10	Argentina	Buenos Aires	41343201	2766890.00
11	American Samoa	Pago Pago	57881	199.00

Showing 1 to 11 of 250 entries, 4 total columns

Console Terminal Background Jobs

R 4.5.2 · ~/Desktop/ITBDA_MapReduce/

```
> # Display the cleaned data and data quality checks
> print(head(clean_df, 10))
               country          capital population area_km2
1              Andorra Andorra la Vella      84000      468
2  United Arab Emirates        Abu Dhabi    4975593    82880
3          Afghanistan            Kabul   29121286   647500
4   Antigua and Barbuda       St. John's      86754      443
5             Anguilla       The Valley      13254      102
6              Albania           Tirana    2986952    28748
7              Armenia          Yerevan    2968000    29800
8               Angola           Luanda   13068161  1246700
9           Antarctica     Not provided          0 14000000
10           Argentina     Buenos Aires   41343201  2766890
> print(colSums(is.na(clean_df)))
   country    capital population   area_km2
         0          0          0          0
>
> cat("Capitals handled:", sum(missing_capital), "\n")
Capitals handled: 8
> cat("Populations imputed:", sum(missing_population), "\n")
Populations imputed: 0
> cat("Areas imputed:", sum(missing_area), "\n")
Areas imputed: 1
> cat("Cleaned records:", nrow(clean_df), "\n")
Cleaned records: 250
```

The above screenshot shows the process of cleaning, preprocessing and denoising the data frame. From the screenshot we can see the data frame now contains cleaned data and all missing values and duplicates of countries were taken care of as well as whitespaces. The screenshot also shows that for missing capitals, populations and area relevant missing value imputation techniques were used.

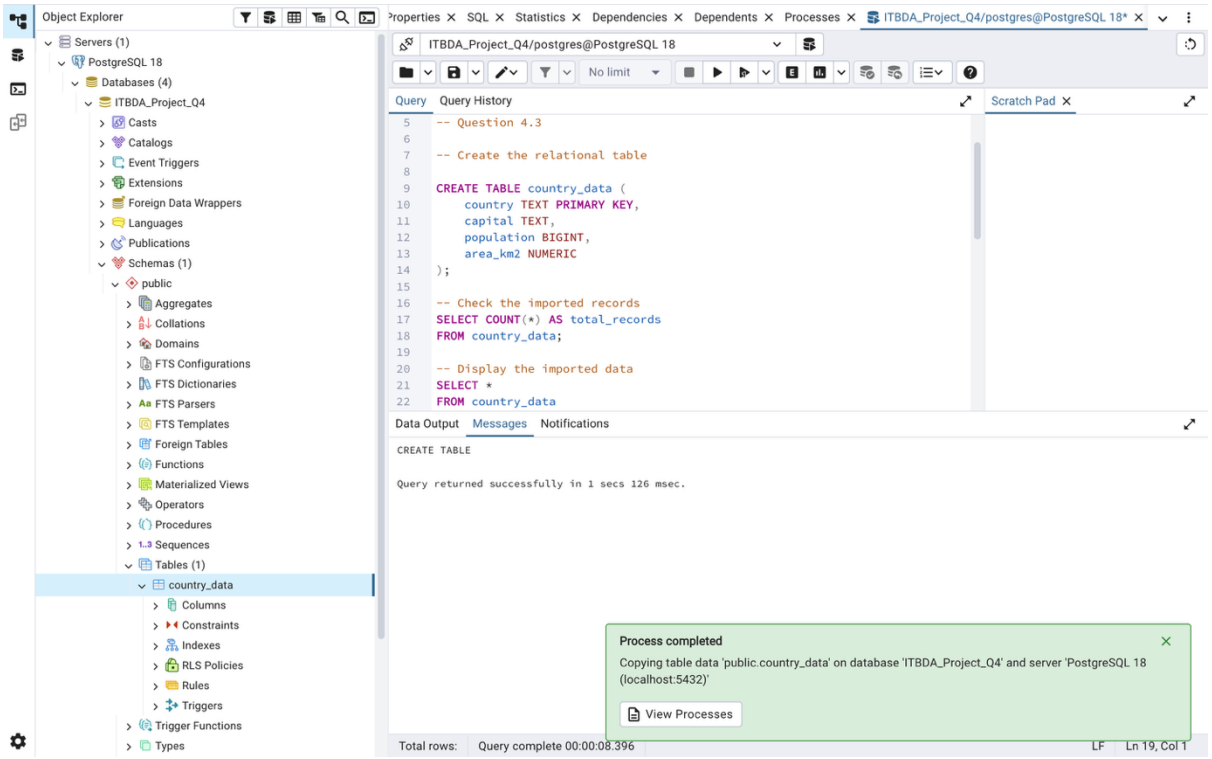
```
106
107  # Export the cleaned dataset without scientific notation for PostgreSQL
108  options(scipen = 999)
109
110  write.csv(clean_df, "/Users/gaby/Desktop/Countries_Clean_Data.csv", row.names = FALSE)
```

110:87 (Top Level) R Sc

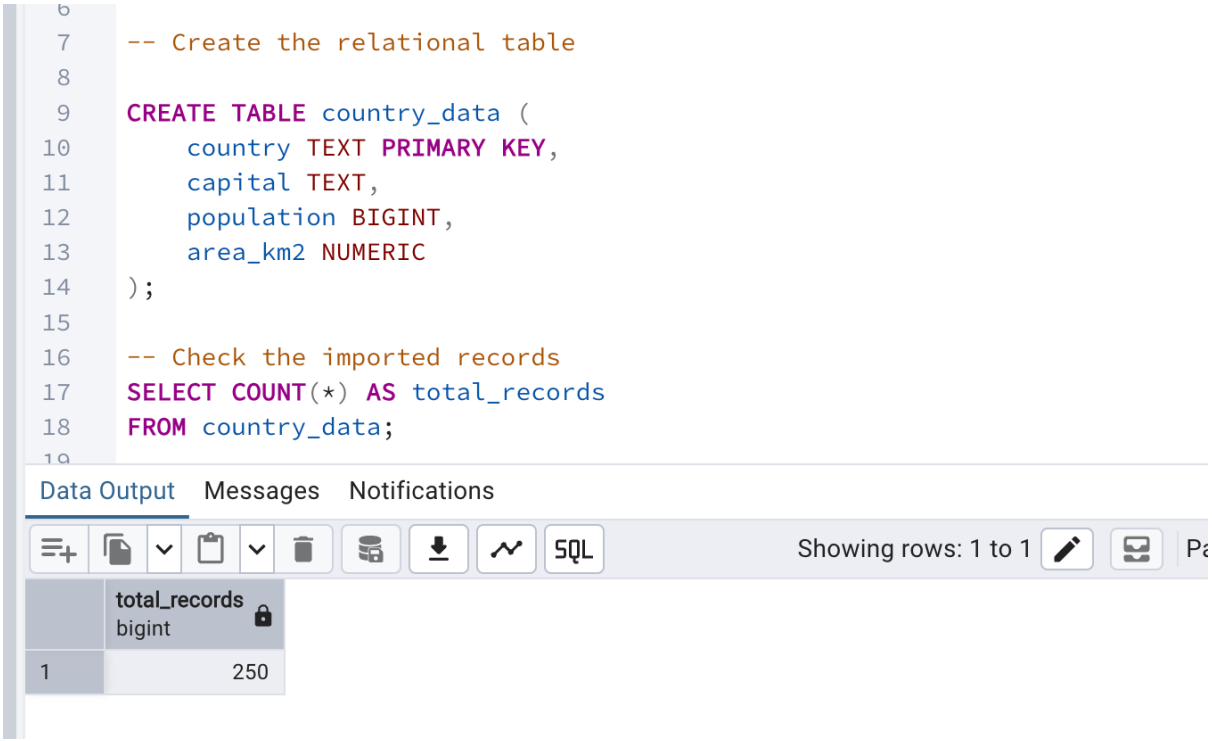
The above screenshot confirms that the cleaned data frame was exported as a csv file: Countries_Clean_Data.csv, which will be used in the next question for PostgreSQL.

QUESTION 4.3

The full code for Question 4.3 is in the SQL file which has been submitted as well. Please refer to the SQL file for the full answer to Question 4.3. (File: Q4PostgreSQL.sql)



The above screenshot shows evidence that the Countries_Clean_Data.csv was imported successfully into the country_data TABLE that was created.



The above screenshot confirms that all 250 records were imported.

```
-- Display the imported data
SELECT *
FROM country_data
LIMIT 10;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10

	country [PK] text	capital text	population bigint	area_km2 numeric
1	Andorra	Andorra la Ve...	84000	468
2	United Arab Emirat...	Abu Dhabi	4975593	82880
3	Afghanistan	Kabul	29121286	647500
4	Antigua and Barbu...	St. John's	86754	443
5	Anguilla	The Valley	13254	102
6	Albania	Tirana	2986952	28748
7	Armenia	Yerevan	2968000	29800
8	Angola	Luanda	13068161	1246700
9	Antarctica	Not provided	0	14000000
10	Argentina	Buenos Aires	41343201	2766890

The above screenshot shows imported data into the country_data TABLE (only showing the top 10 records, purely for confirmation).

```sql
-- Investigation 1
-- Largest Countries and Capitals by Area

SELECT
    country,
    capital,
    area_km2,
    population
FROM country_data
WHERE area_km2 > 0
ORDER BY area_km2 DESC
LIMIT 10;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page

	country [PK] text	capital text	area_km2 numeric	population bigint
1	Russia	Moscow	17100000	140702000
2	Antarctica	Not provided	14000000	0
3	Canada	Ottawa	9984670	33679000
4	United Stat…	Washington	9629091	310232863
5	China	Beijing	9596960	1330044000
6	Brazil	Brasília	8511965	201103330
7	Australia	Canberra	7686850	21515754
8	India	New Delhi	3287590	1173108018
9	Argentina	Buenos Air…	2766890	41343201
10	Kazakhstan	Astana	2717300	15340000

The above screenshot shows the first SQL query that was run in order to investigate the largest countries and their respective capitals. The data is ordered in descending order and only the first 10 records are shown for ease. We can see that the largest countries are Russia, Antarctica and Canada.

```sql
-- Investigation 2
-- Highest Population Density

SELECT
    country,
    population,
    area_km2,
    ROUND(population / area_km2, 2) AS population_density
FROM country_data
WHERE population >= 100000
  AND area_km2 > 0
ORDER BY population_density DESC
LIMIT 10;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page N

	country [PK] text	population bigint	area_km2 numeric	population_density numeric
1	Singapore	4701069	692.7	6786.59
2	Hong Kong	6898686	1092	6317.48
3	Macao	449198	254	1768.50
4	Maldives	395650	300	1318.83
5	Malta	403000	316	1275.32
6	Bahrain	738004	665	1109.78
7	Banglade...	156118464	144000	1084.16
8	Barbados	285653	431	662.77
9	Palestine	3800000	5970	636.52
10	Taiwan	22894384	35980	636.31

The above screenshot shows the second SQL query that was run in order to investigate the countries with the highest population density (population / areakm2). We used countries where the population was greater than 100 000 just to help the query run a bit faster. The data is ordered in descending order and only the first 10 records are shown for ease. We can see by the above that the top 3 countries with the highest population density are Singapore, Hong Kong and Macao.

From the two investigations above, we can see that the larger the areakm2 of country does not necessarily mean that the country has a high population density.

QUESTION 4.4

GitHub Link: https://github.com/gabrielladelgadox/ITBDA-Project2.git

All the project documentation, notebooks and datasets are included.

Reference List

EMC Education Services. (2015) Data Science & Big Data Analytics: Discovering, Analyzing, Visualizing, and Presenting Data. Indianapolis, IN: John Wiley & Sons.

AI Evidence Section

The following is a link for the AI chat for evidence of AI used in the project:

https://chatgpt.com/share/6af4be95-0858-83ea-9694-3869f52d0129

AI Disclosure Appendix

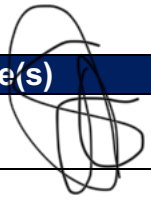
AI tool used	Purpose / how it was used	How I verified or adapted the output
ChatGPT	To check grammar and language	Confirmed recommendations were valid and made the necessary adjustments

Declaration:

I/we declare that this assessment is my/our own original work, except for source material explicitly acknowledged. I/we confirm that any use of Artificial Intelligence (AI) tools has been applied only as permitted by the assessment instructions and has been fully disclosed where required.

I/we understand that I/we remain responsible for the accuracy, originality, and academic quality of this submission. I/we am/are aware of the institution's policies on academic integrity and the requirements relating to the responsible use of AI in assessments.

I/we acknowledge that failure to comply with these requirements, including undeclared or inappropriate AI use, may result in academic penalties or disciplinary procedures in accordance with institutional policy.

Student Signature(s)	Date
	11 November 2026